

# Younsoo Park

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## Education

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**The Pennsylvania State University**

University Park, PA

Bachelor of Science in Computer Science Minor in Mathematics Expected Graduation: May 2027 Dean's List

## Research Interests

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Systems Security  
Network Security  
Distributed Systems  
LLM Systems  
AI Agent Memory  
Federated Learning  
TinyML for IoT Security  
Adversarial Machine Learning  
Edge AI Systems

## Research Experience

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**Undergraduate Research Assistant & Team Lead**

Oct 2025 – Present

*Reliable Federated TinyML Deployment for IoT Security (WIP, LCTES '26)*

Penn State University and Elizabethtown College Supervisors: Dr. Suman Saha and Dr. Peilong Li

Research focus: Federated and adversarially robust TinyML systems for IoT security and resource-constrained edge-device deployment.

Key contributions:

- Designed a federated learning pipeline using FedAvgM, focal loss, and cosine learning-rate scheduling on CIC-IDS2017.
- Improved attack recall from 46.7% to 93.85% and F1-score from 84.1% to 89.32%.
- Built a TinyML compression pipeline including BatchNorm folding, structured pruning, knowledge distillation, and INT8 QAT/PTQ.
- Achieved  $12.28\times$  model compression and 74.5% inference latency reduction for ESP32 deployment.
- Evaluated adversarial robustness using PGD attacks ( $\epsilon$ -sweep 0.01–0.2) across 48 compression configurations.

## Publications and Manuscripts

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**Reliable Federated TinyML Deployment for IoT Security** Work in Progress Target Venue: ACM LCTES 2026

## Independent Study

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**AI Agent Memory Systems**

Jul 2026 – Present

Advisor: Prof. Kiwan Maeng

- Researched five production and research AI agent memory systems (Mem0, A-MEM, LangMem, Zep, OpenAI Memory) to identify the SOTA approach, comparing memory creation, retrieval, and lifecycle-management approaches.
- Identified a core architectural divergence: Mem0 extracts and retrieves discrete facts, while A-MEM models memory as an evolving, interconnected knowledge structure; traced how Mem0's open-source implementation diverged from its published ADD/UPDATE/DELETE design into an ADD-only, retrieval-time resolution approach.
- Narrowed research focus to Mem0 (implementation depth, practical extensibility) and A-MEM (architectural novelty) for application to the lab's production AI agent system.
- Applying memory architectures to production recommendation systems through a Memory-first Personalization framework, shipped as Wowness at Levit Inc.

## Industry and Technical Experience

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### Levit Inc. (Shoppport AI)

Seoul, South Korea

Associate Problem Solver

June 2026 – Aug 2026

- Worked directly with the CEO to define product direction, prioritize engineering initiatives, and translate customer problems into production AI systems for an LLM-powered shopping assistant.
- Built a production evaluation platform analyzing **49,807 conversations** and **80,124 query-level routing decisions**, achieving **97.7% evaluation coverage** through versioned routing audit and historical backfill pipelines that reprocessed 47,231 conversations spanning 100+ days of production history.
- Developed end-to-end experimentation infrastructure across frontend, backend, workers, and admin systems, improving recommendation completion from **62.8%** to **75.0% (+12.2 percentage points)**.
- Evolved Shoppport's routing stack through three architectural generations – LLM classification, an embedding-based semantic fast path ( $\sim 8\times$  faster for confident cases), and a hybrid classifier for ambiguous queries – reaching **86% V3 routing accuracy** and rescuing **93%** of eligible router errors.
- Designed and shipped Wowness, a Mem0-backed cross-category discovery recommender grounded in the live catalog, as sole engineer across a 919-line NestJS engine, React UI, and admin review lab – extending the Memory-first Personalization architecture from proposal to production.
- Diagnosed production failures at the system boundary, including a split MongoDB connection that held the V3 feature switch at zero executions, and corrected systematic analytics false positives, reducing one queue-level count from 2,005 to 811.
- Contributed **730 GitHub commits** and **143 merged pull requests**, adding over **114,000 lines** of TypeScript/TSX during the engagement.

### SIGNUM (Nittany AI Alliance – PIT-UN Funded)

Aug 2024 – Nov 2025

Co-founder of a \$3,750 PIT-UN funded AI platform for caregiver decision support.

- Built multi-source ETL pipelines integrating CMS, NPPES, and Google Places datasets into a DuckDB warehouse.
- Developed hospital-rating prediction models using Markov Transition analysis with confidence intervals.
- Designed and deployed domain-specific RAG systems using FAISS and AWS Bedrock, reducing caregiver research time from 50+ hours per week to near-instant answers.

**Penn State Advanced Vehicle Team (EcoCAR)**

University Park, PA

Project Manager

Aug 2026 – Present

- Manage planning and deliverables for Penn State's team in the EcoCAR Challenge, managed by Argonne National Laboratory with GM and MathWorks as sponsors.
- Building the project structure and processes in Year 1 of the competition that future teams will build on, while keeping vehicle work on schedule.

**Atom Tech Solutions LTD**

Berkeley, CA

Software Engineering Intern

May 2022 – Aug 2022

- Built secure authentication systems using SQL/JWT authentication and encrypted credential storage.
- Implemented automated credential generation and authenticated onboarding workflows.

**Technical Projects**

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**GladIEEEators – Penn State IEEE Battle Bots (1st Place)**

Apr 2026

- Designed Shot & Chaser, a coordinated two-robot combat system, for the IEEE Student Chapter Battle Bots Competition, winning 1st place.
- Engineered a belt-driven AR500 vertical-spinner weapon system (334 J at 18,500 RPM, 200+ mph tip speed), validated via Fusion FEA above 4,000 N with a 2.09 minimum safety factor.
- Designed mixed-material armor (6061 aluminum, UHMW, TPU) and integrated drivetrain, radio control, and protected LiPo electronics across both robots within a \$976 total build budget.

**XIIO – LinkedIn for Filmmakers**

Apr 2026 – Present

- Built a creator networking platform featuring portfolios, video streaming, collaboration workflows, and school-based discovery using Next.js, Firebase, and Cloudflare Stream.
- Designed real-time messaging, creator profiles, media uploads, and production portfolio management.

**ASME @ Penn State Website**

Aug 2024 – Present

- Developed the official ASME Penn State website using React, TypeScript, Vite, and Firebase.
- Designed a multi-tier permission and approval system for organization management.

**Hangukwon (Family Business Web App)**

2024 – Present

- Developed a multilingual full-stack restaurant web application using React, TypeScript, Node.js/Express, and MongoDB Atlas.
- Implemented JWT authentication, Google Maps API integration, and multilingual i18n support.

**Teaching Experience**

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**Lead Teaching Assistant, Math 230 (Multivariable Calculus)** Penn State University

Aug 2025 – Present

Responsibilities include assisting instruction, supporting students during office hours, and helping coordinate course activities.

## Leadership and Service

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|------------------------------------------------------------------------|-----------------------|
| <b>THON Technology Committee</b> Technology Captain (2027)             | Apr 2026 – Present    |
| <b>Association for Computing Machinery (ACM)</b> Web & Design Director | Apr 2026 – Present    |
| <b>Republic of Korea Air Force</b> Squad Leader                        | Sept 2022 – June 2024 |

## Relevant Coursework

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### Computer Science

- CMPSC 131: Programming and Computation I
- CMPSC 132: Programming and Computation II
- CMPSC 221: Object Oriented Programming
- CMPSC 311: Introduction to Systems Programming
- CMPSC 360: Discrete Mathematics
- CMPSC 461: Programming Language Concepts
- CMPSC 464: Theory of Computation
- CMPSC 465: Data Structures and Algorithms

### Computer Engineering

- CMPEN 270: Digital Design
- CMPEN 331: Computer Organization and Design

## Awards and Honors

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|---------------------------------------------------------|----------|
| 1st Place, ThinkNeuro LLC Project Competition           | Apr 2026 |
| 1st Place, IEEE Student Chapter Battle Bots Competition | Apr 2026 |
| PIT-UN Funding Award (\$3,750)                          | Apr 2025 |
| 3rd Place, ACM Pathfinder Robotics Challenge            | Apr 2025 |
| Republic of Korea Air Force Commendation Award          | Aug 2023 |

## Technical Skills

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**Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript, SQL

**Frameworks:** React, Next.js, Node.js, NestJS, Express, Flask, Firebase

**Machine Learning:** PyTorch, TensorFlow, Flower, Scikit-learn, spaCy, SHAP, FAISS, Mem0, AWS Bedrock, OpenAI API

**Data and DevOps:** PostgreSQL, MongoDB, Redis, DuckDB, Docker, Kubernetes, Helm, GitHub Actions, Datadog, Git